

Experiment-19

Aim

To perform edge detection along both X-axis and Y-axis using the Sobel operator and obtain the combined edge image.

Procedure

1. Select an input image.
2. Read the image using OpenCV.
3. Convert the input image into grayscale.
4. Apply the Sobel operator along the X-axis.
5. Apply the Sobel operator along the Y-axis.
6. Combine the X and Y gradients to obtain the XY edge image.
7. Convert the result into an 8-bit image.
8. Display the original image and the final XY edge-detected image.
9. Observe the edges detected in both horizontal and vertical directions.

Python Code

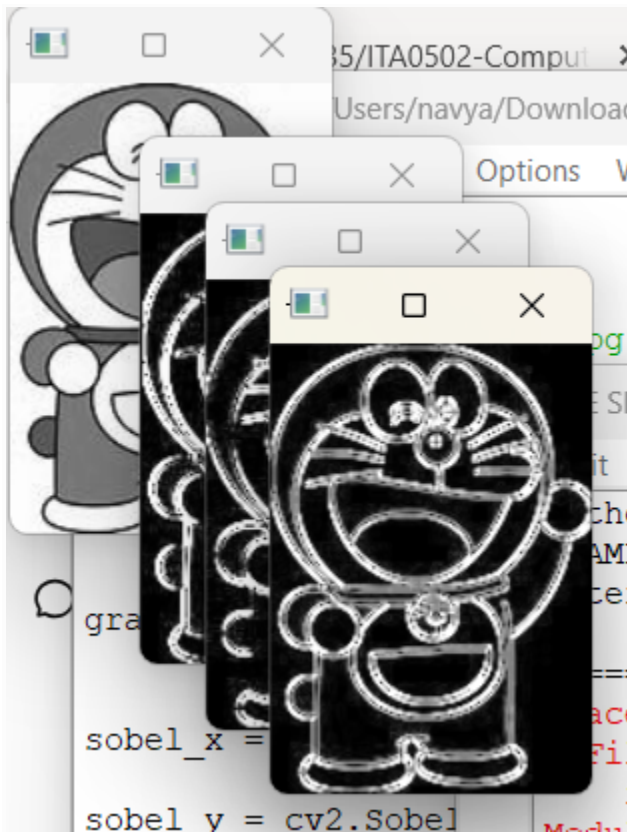
```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)
sobel_xy = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)
cv2.imshow("Original Image", gray)
cv2.imshow("Sobel X", sobel_x)
```

```
cv2.imshow("Sobel Y", sobel_y)
cv2.imshow("Sobel XY Edge Detection", sobel_xy)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The Sobel operator was successfully applied along both the X-axis and Y-axis. The combined XY Sobel edge image successfully detects horizontal and vertical edges present in the input image.